

# Exercise Sheet 5

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## Bayesian Networks

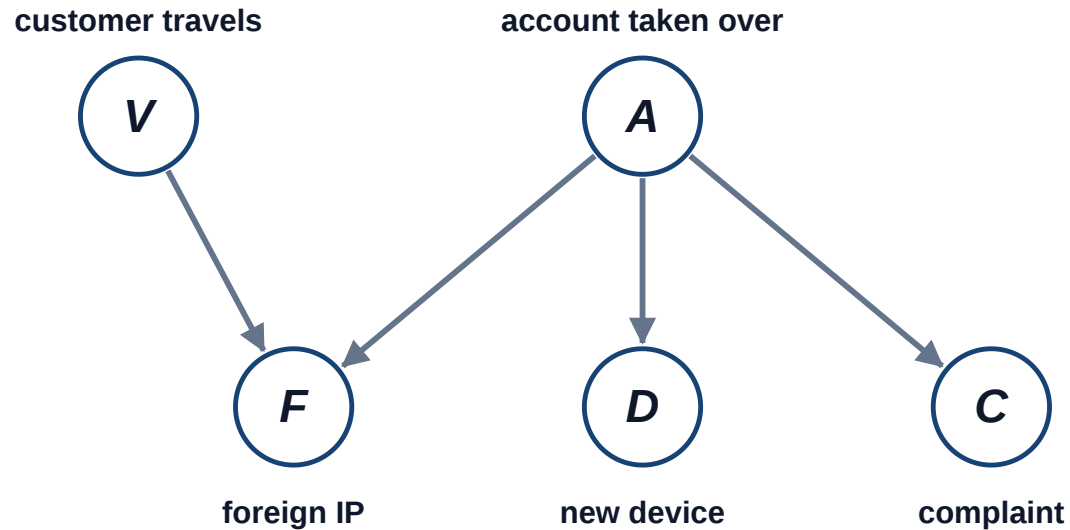
Probabilistic Machine Learning

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# 5.A Graphing a Joint

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# Exercise 5.A.1: A Small Fraud Network



All five variables are yes/no.

1. Write the joint  $p(V, A, F, D, C)$  as the network factorises it.
2. How many numbers does the network need? How many the full joint?
3. State the local Markov claim for  $D$  and for  $F$ .
4. Name one missing arrow you would challenge, and what adding it costs.

# Exercise 5.A.1: A Small Fraud Network — Solution

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1.  $p(V, A, F, D, C) = p(V) \cdot p(A) \cdot p(F \mid V, A) \cdot p(D \mid A) \cdot p(C \mid A)$ .
2.  $1 + 1 + 4 + 2 + 2 = 10$  numbers, against  $2^5 - 1 = 31$ .
3.  $D$  has parent  $A$  and no descendants:  $D \perp\!\!\!\perp \{V, F, C\} \mid A$ .  $F$  has parents  $V, A$ :  $F \perp\!\!\!\perp \{D, C\} \mid V, A$ .
4. For example  $V \rightarrow D$ : travellers often use a new phone or a hotel computer. It removes the claim  $D \perp\!\!\!\perp V \mid A$  and turns  $p(D \mid A)$  into  $p(D \mid V, A)$ : 4 numbers instead of 2, 12 in total. (Any well-argued arrow counts; the point is that each missing arrow is a claim.)

## Exercise 5.A.2: Inference by Elimination

Tables:  $P(V=1) = 0.05$ ,  $P(A=1) = 0.01$ ;  $P(F=1 \mid V, A)$ : 0.02 for (0, 0), 0.9 for (1, 0), 0.7 for (0, 1), 0.95 for (1, 1);  $P(D=1 \mid A)$ : 0.1 for  $A=0$ , 0.6 for  $A=1$ ;  $P(C=1 \mid A)$ : 0.001 for  $A=0$ , 0.5 for  $A=1$ .

1. Compute  $P(F=1)$ . Which variables can be dropped before summing, and why? Eliminate  $V$  first and give the message  $m(A)$ .
2. Compute  $P(A=1 \mid F=1)$ .
3. Compute  $P(A=1 \mid F=1, D=1)$ .
4. Compute  $P(A=1 \mid F=1, V=1)$  and compare with 2. Name the effect.

## Exercise 5.A.2: Inference by Elimination — Solution

## Exercise 5.A.2: Inference by Elimination — Solution

1.  $D$  and  $C$  are unobserved leaves:  $\sum_d p(d \mid A) = 1$  and  $\sum_c p(c \mid A) = 1$ , so they drop out. Then  $P(F=1) = \sum_A p(A)m(A)$  with  $m(A) = \sum_V p(V)P(F=1 \mid V, A)$ :  $m(0) = 0.95 \cdot 0.02 + 0.05 \cdot 0.9 = 0.064$ ,  $m(1) = 0.95 \cdot 0.7 + 0.05 \cdot 0.95 = 0.7125$ .  $P(F=1) = 0.99 \cdot 0.064 + 0.01 \cdot 0.7125 = 0.06336 + 0.007125 = 0.070485$ .
2.  $P(A=1 \mid F=1) = 0.007125/0.070485 \approx 0.101$  — ten times the prior.
3. Multiply in  $P(D=1 \mid A)$ :  $0.007125 \cdot 0.6 = 0.004275$  and  $0.06336 \cdot 0.1 = 0.006336$ , so  $P(A=1 \mid F=1, D=1) = 0.004275/0.010611 \approx 0.403$ .
4.  $P(A=1, V=1, F=1) = 0.01 \cdot 0.05 \cdot 0.95 = 0.000475$  and  $P(A=0, V=1, F=1) = 0.99 \cdot 0.05 \cdot 0.9 = 0.04455$ , so  $P(A=1 \mid F=1, V=1) = 0.000475/0.045025 \approx 0.011$  — back to about the prior. Travel **explains away** the foreign IP:  $V$  and  $A$  are independent, but not given their common effect  $F$ .

# 5.B Reading Independence

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## Exercise 5.B.1: d-Separation

For the fraud network of 5.A.1, decide with d-separation — give the path and the triple that blocks or opens it:

1.  $V \perp\!\!\!\perp A$ ?
2.  $V \perp\!\!\!\perp A \mid F$ ?
3.  $V \perp\!\!\!\perp D$ ?
4.  $V \perp\!\!\!\perp D \mid F$ ?
5.  $V \perp\!\!\!\perp D \mid F, A$ ?
6.  $D \perp\!\!\!\perp C$ ? and  $D \perp\!\!\!\perp C \mid A$ ?

Then check 3 and 4 with the numbers of 5.A.2: compare  $P(V=1 \mid F=1)$  with  $P(V=1 \mid F=1, D=1)$ .

# Exercise 5.B.1: d-Separation — Solution

## Exercise 5.B.1: d-Separation — Solution

| query                            | path                                                                   | verdict |
|----------------------------------|------------------------------------------------------------------------|---------|
| $V \perp\!\!\!\perp A$           | $V \rightarrow F \leftarrow A$ : collider, unobserved                  | yes     |
| $V \perp\!\!\!\perp A \mid F$    | the collider is observed: open                                         | no      |
| $V \perp\!\!\!\perp D$           | $V \rightarrow F \leftarrow A \rightarrow D$ : collider $F$ unobserved | yes     |
| $V \perp\!\!\!\perp D \mid F$    | collider $F$ open, fork $A$ unobserved: open                           | no      |
| $V \perp\!\!\!\perp D \mid F, A$ | fork $A$ observed: blocked                                             | yes     |
| $D \perp\!\!\!\perp C$           | $D \leftarrow A \rightarrow C$ : fork, unobserved                      | no      |
| $D \perp\!\!\!\perp C \mid A$    | fork observed                                                          | yes     |

**Numbers.**  $P(V=1) = 0.05 = P(V=1 \mid D=1)$  (claim 3). Given  $F=1$ :  $P(V=1 \mid F=1) = 0.045025/0.070485 \approx 0.639$ , but  $P(V=1 \mid F=1, D=1) = 0.00474/0.010611 \approx 0.447$ : a new device makes a take-over more likely, which explains the foreign IP and makes travel less likely. Claim 4 fails, as d-separation said.

## Exercise 5.B.2: Which Arrows Are in the Data?

1. Which of these graphs are Markov equivalent? (a)  $X \rightarrow Y \rightarrow Z$  · (b)  $X \leftarrow Y \leftarrow Z$  · (c)  $X \leftarrow Y \rightarrow Z$  · (d)  $X \rightarrow Y \leftarrow Z$  · (e) (a) plus  $X \rightarrow Z$
2. In the fraud network of 5.A.1, which single arrows can be reversed without changing any independence claim? Can  $A \rightarrow D$  and  $A \rightarrow C$  be reversed together?
3. An analyst fits the fraud network from data and says "take-overs **cause** new devices". What can the data support?

## Exercise 5.B.2: Which Arrows Are in the Data? — Solution

## Exercise 5.B.2: Which Arrows Are in the Data? — Solution

1. (a), (b), (c) share the skeleton  $X - Y - Z$  and have no collider: all claim  $X \perp\!\!\!\perp Z \mid Y$  and nothing else — **equivalent**. (d) has the collider at  $Y$ : it claims  $X \perp\!\!\!\perp Z$  instead — different. (e) is complete and claims no independence at all — different from all four.
2. Equivalent graphs need the same skeleton and the same colliders. The collider  $V \rightarrow F \leftarrow A$  must stay, so  $V \rightarrow F$  and  $A \rightarrow F$  cannot be reversed.  $A \rightarrow D$  alone can be reversed ( $D \rightarrow A \rightarrow C$  creates no collider), and so can  $A \rightarrow C$  alone. Both together give  $D \rightarrow A \leftarrow C$ , a new collider — not equivalent.
3. Only that  $A$  and  $D$  are dependent, and — since  $A \rightarrow D$  is reversible — not its direction. The causal reading comes from knowledge outside the data (a take-over happens before the device is used), not from the fit.

# 5.C Plates and Learning

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# Exercise 5.C.1: Naive Bayes as a Plate Diagram

Naive Bayes for  $N$  transactions with  $p$  yes/no flags: class prior  $\pi = P(y = 1)$  with a  $\text{Beta}(a, b)$  prior; flag probabilities  $\theta_{jc} = P(x_j = 1 \mid y = c)$ , each with a  $\text{Beta}(1, 1)$  prior.

1. Draw the plate diagram: nodes, shading, dots, plates (and which plates are nested).
2. Write the factorisation that the diagram encodes.
3. At prediction time a new transaction arrives with flags but no label. What changes in the drawing?

# Exercise 5.C.1: Naive Bayes as a Plate Diagram — Solution

# Exercise 5.C.1: Naive Bayes as a Plate Diagram — Solution

1. Dot  $(a, b) \rightarrow$  open node  $\pi \rightarrow$  shaded  $y_i$ . A plate over  $N$  transactions holds  $y_i$  and, **nested** inside it, a plate over the  $p$  flags holding the shaded  $x_{ij}$ . Outside the  $N$  plate: open nodes  $\theta_{jc}$  in a plate over the  $p$  flags and a plate over the 2 classes, fed by a dot  $(1, 1)$ . Arrows  $y_i \rightarrow x_{ij}$  and  $\theta_{jc} \rightarrow x_{ij}$  ( $y_i$  selects which  $\theta_j$  applies).

2.

$$p(\pi \mid a, b) \prod_{j=1}^p \prod_{c=0}^1 p(\theta_{jc}) \prod_{i=1}^N \left[ p(y_i \mid \pi) \prod_{j=1}^p p(x_{ij} \mid y_i, \theta_{j0}, \theta_{j1}) \right]$$

3. The new label  $y_{\text{new}}$  is **unshaded** — it is inferred. The parameters are either conditioned on the training data (the full Bayesian answer) or replaced by their smoothed counts — the posterior means of Lecture 04.

# Exercise 5.C.2: Counting, Elimination, and an Empty Row

1. A chain  $X_1 \rightarrow \dots \rightarrow X_{10}$  of yes/no variables. How many terms does  $p(x_{10})$  need if all other variables are summed at once, and roughly how many operations by elimination?
2. From 200 records of the fraud network (table), estimate  $P(F=1 \mid V, A)$  by maximum likelihood and with one pseudo-count per value. What happens to the last row, and what does smoothing put there?

| $V$ | $A$ | records | with $F = 1$ |
|-----|-----|---------|--------------|
| 0   | 0   | 180     | 4            |
| 1   | 0   | 15      | 13           |
| 0   | 1   | 5       | 3            |
| 1   | 1   | 0       | 0            |

## Exercise 5.C.2: Counting, Elimination, and an Empty Row — Solution

## Exercise 5.C.2: Counting, Elimination, and an Empty Row — Solution

1.  $2^9 = 512$  terms (each a product of 10 factors), against about  $4 \cdot 9 = 36$  multiplications by passing a two-entry message along the chain.

2.

| $V, A$ | MLE                   | smoothed $(n_1 + 1)/(n + 2)$ |
|--------|-----------------------|------------------------------|
| 0, 0   | $4/180 \approx 0.022$ | $5/182 \approx 0.027$        |
| 1, 0   | $13/15 \approx 0.867$ | $14/17 \approx 0.824$        |
| 0, 1   | $3/5 = 0.6$           | $4/7 \approx 0.571$          |
| 1, 1   | $0/0$ — undefined     | $1/2 = 0.5$                  |

A parent configuration that never occurred leaves its row **undetermined** under maximum likelihood; the pseudo-counts fill it with the prior mean  $1/2$ . Sparse rows are one more reason to keep the number of parents small.